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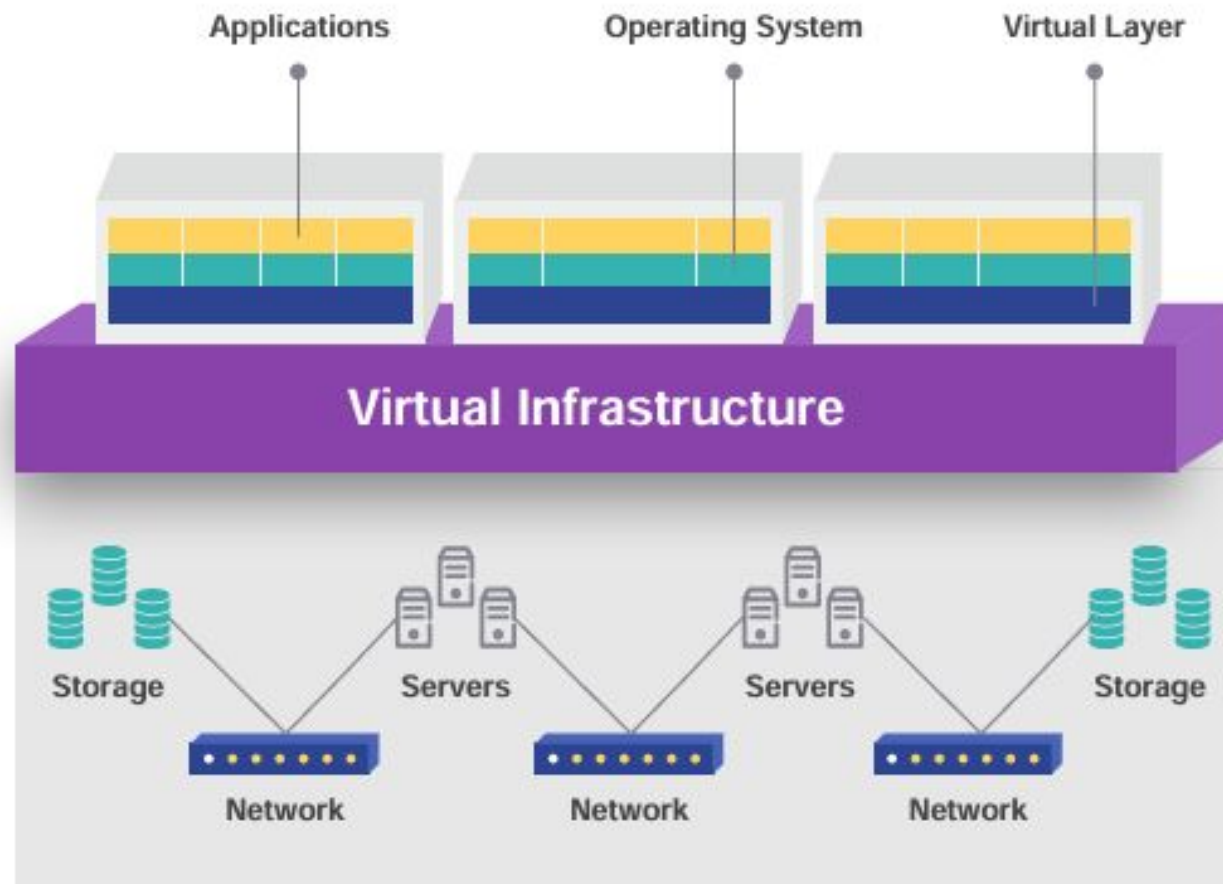
- ✓ What is virtual hard disk and its usage in a different virtualized environment

04 Hypervisor

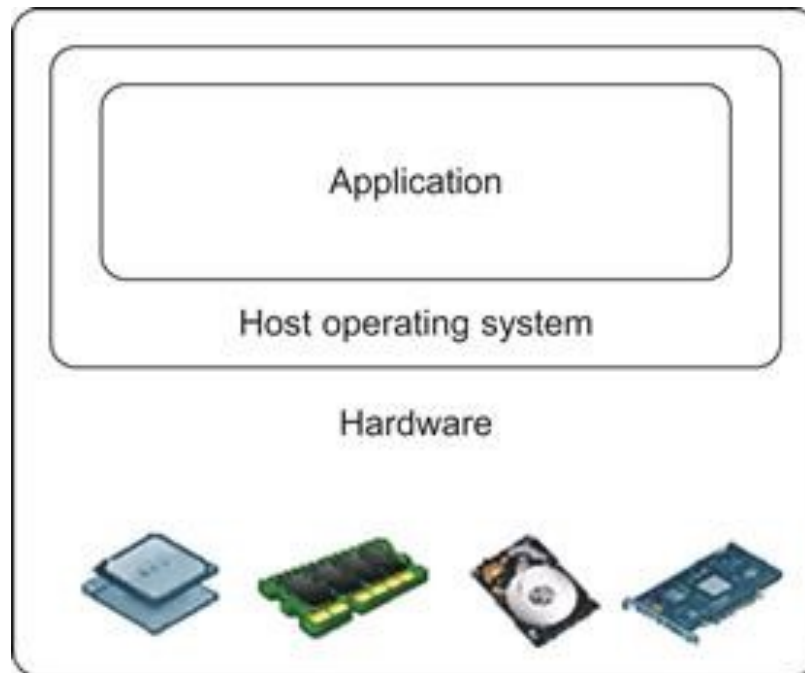
- ✓ Different types of Hypervisor management application.



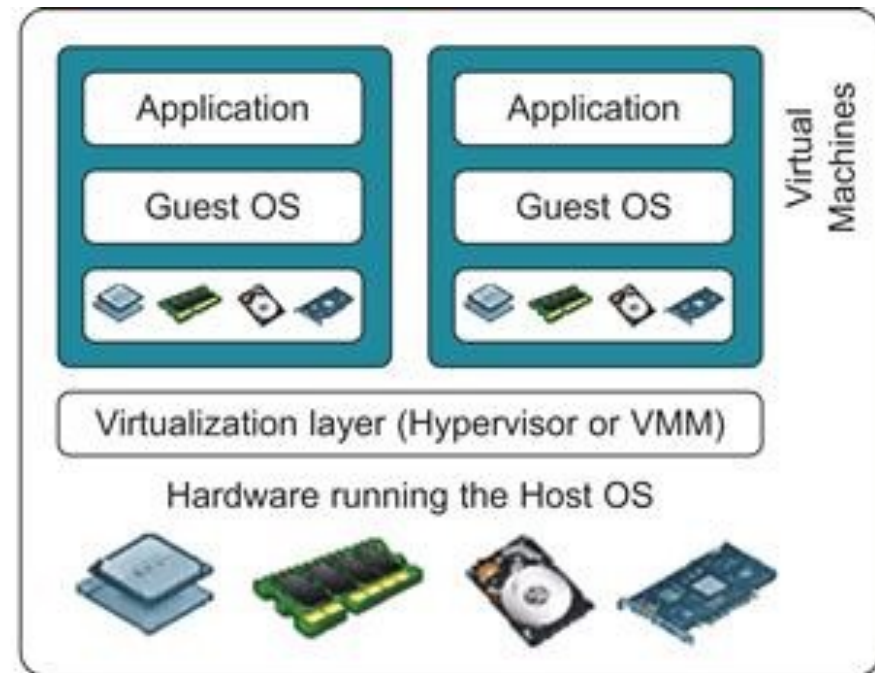
Recap – Virtualization



Need of Virtualization

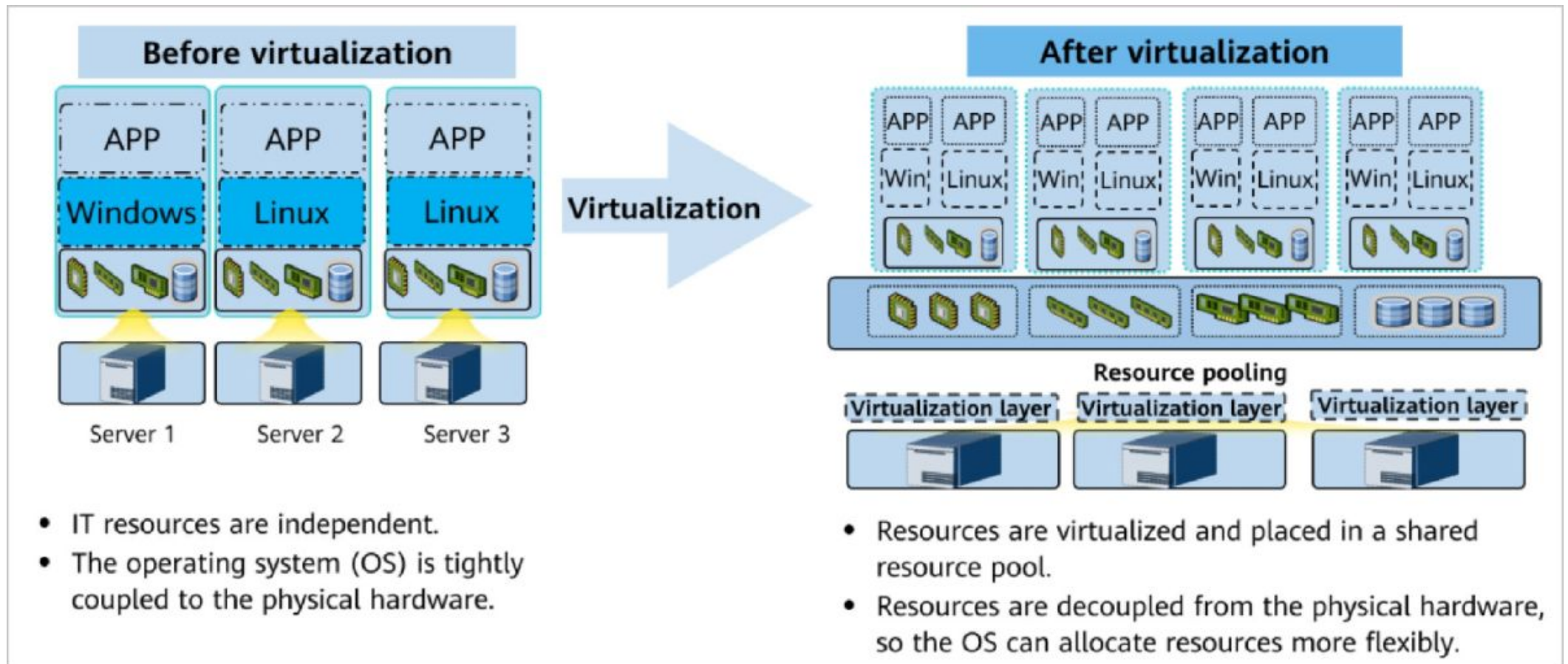


(a) Traditional computer



(b) After virtualization

Virtualization vs. Non- Virtualization



Understanding Virtualization

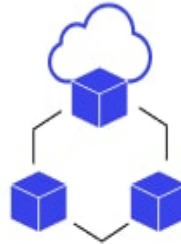
- **Virtual machine manager (VMM):** A VMM, also called the hypervisor, is software that creates and manages virtual machines. It is a specialized operating system that uses minimal hardware resources so that memory and processing are available for the VMs it creates.
- **Virtual machine (VM):** A VM is a machine created by a hypervisor/VMM that runs like any other computer. It usually needs an operating system installed on it to become functional. A VM uses the VMM/hypervisor for access to memory, CPU, network, video, and other resources.

Virtualization Technologies

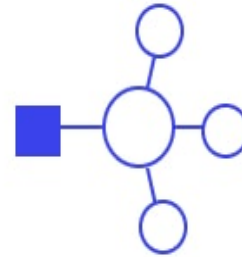
Types of Virtualization



Desktop



Data



Network



Storage



Server

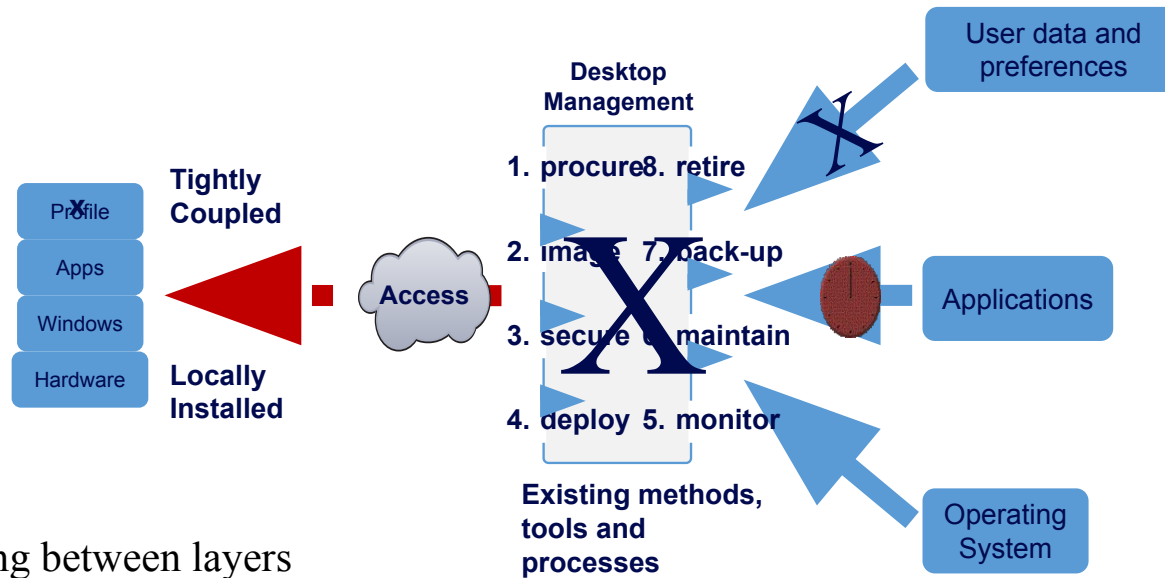


Application



Cloud

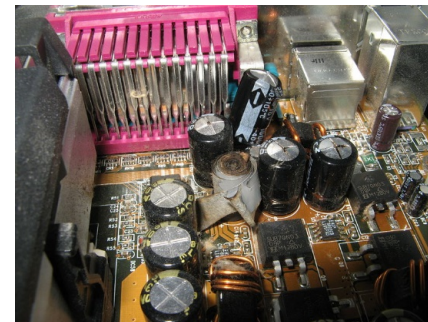
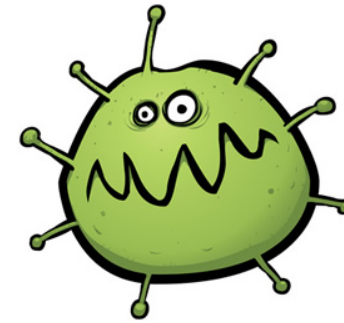
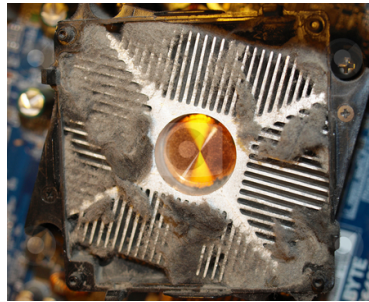
Traditional Desktop Deployment



- Tight binding between layers
- The components are linked together in ways that are difficult to support and maintain
- A problem at one layer often causes a chain reaction
- May destroy the whole stack
- Makes recovery difficult
- Threatens any locally stored user data and settings
- Most organisations just replace or re-image the whole PC

Need of Virtualization Desktop Infrastructure

Necessity of Virtual Desktop Infrastructure



Virtualization Desktop Infrastructure

VDI

Virtual desktop infrastructure (VDI) is the creation and management of virtual desktops and applications that allow employees to access and work on these applications and services while they are in the office, outside the office, or even from a remote location.

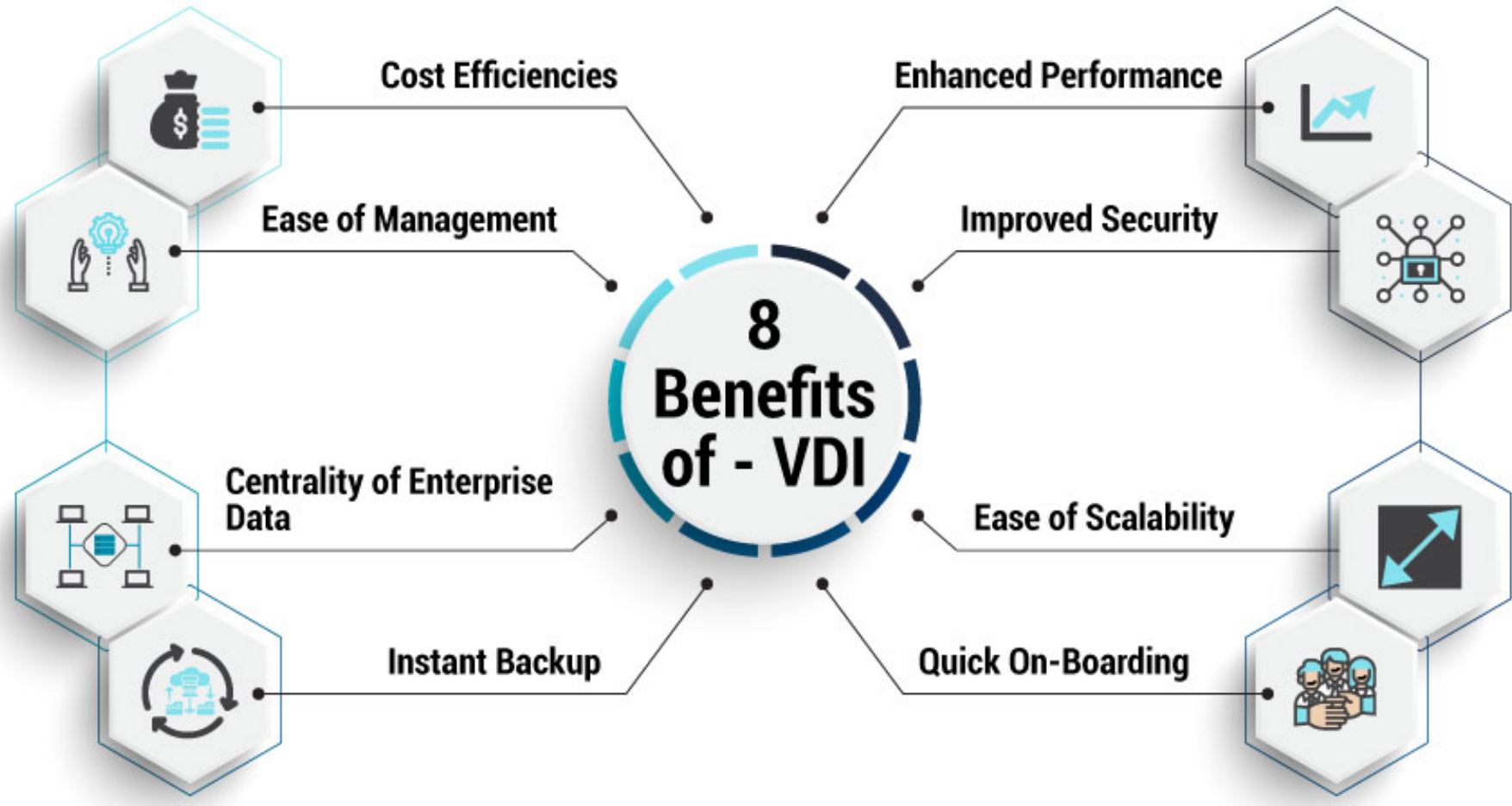


Virtualization Desktop Infrastructure

- Virtual Desktop Infrastructure in short VDI refers to the use of virtual servers to supply and manage virtual desktops.
- VDI centralizes operating systems and makes them accessible to end users on command.
- VDI is a desktop virtual server that runs and controls a desktop environment, often Microsoft Windows, inside a data center.



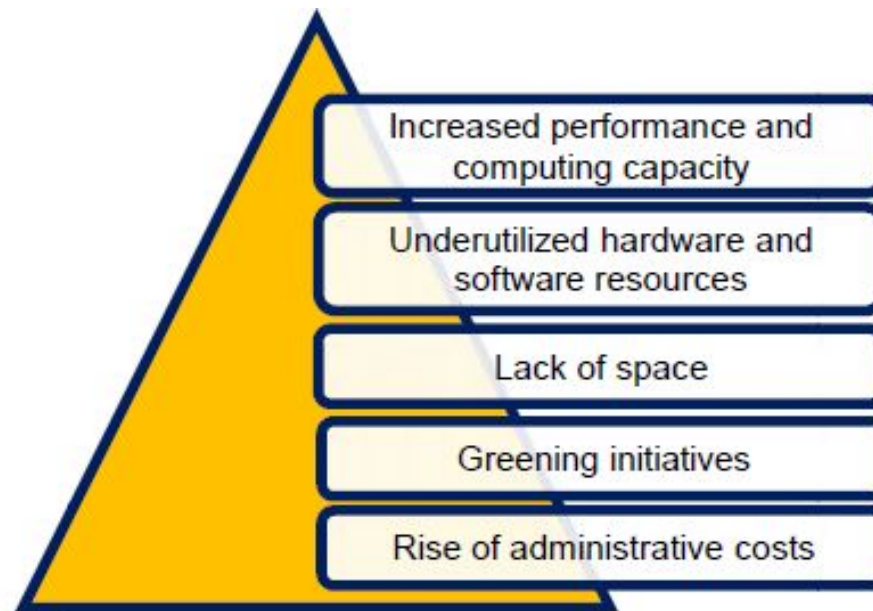
Benefits of VDI



Virtual Machine – Introduction

- A VM involves an isolated guest OS installation within a normal host OS.
- From the user perspective, VM is software platform like physical computer that runs OSs and apps.
- VMs possess hardware virtually.

Driving Factors of Virtual Machines



Virtual Machines Components

- It has the following major components
- Virtual machines typically have an **Operating system, VMware Tools, Virtual resources and hardware.**

Setup a Virtual Machine

Step 1: Make a blank virtual machine.

Step 2: Create a virtual disc for storage.

Step 3: Add a network interface to link the virtual machine to the network.

Step 4: Install the guest operating system (OS).

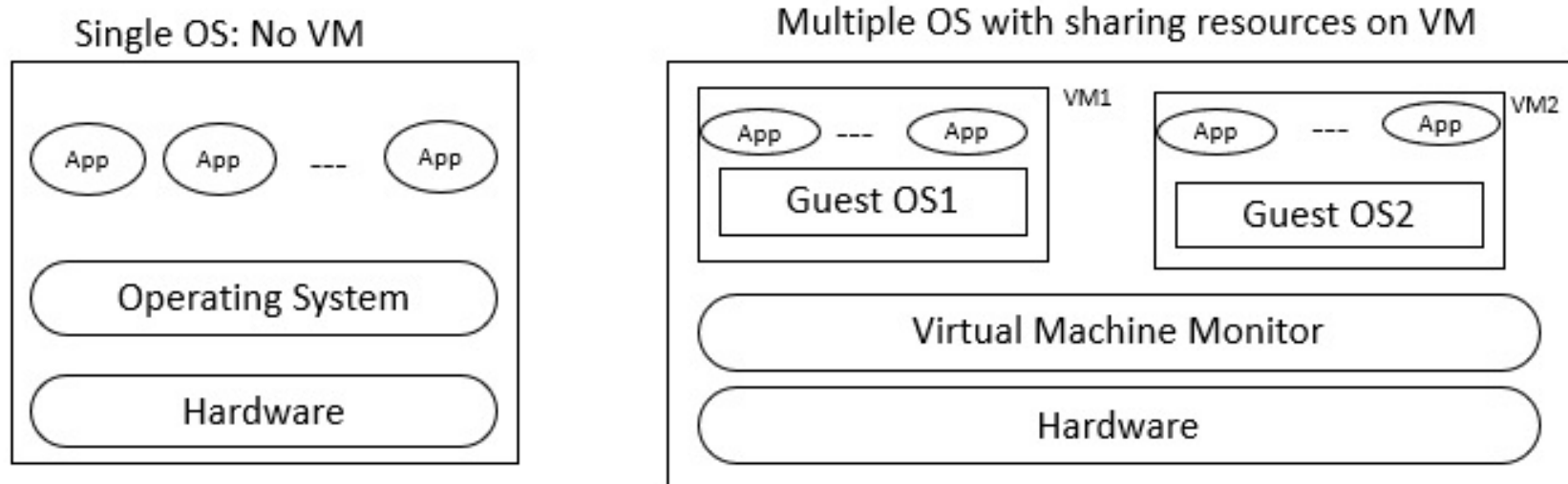
Step 5: Register the VM with the content delivery network and attach any necessary subscriptions.

Step 6: Install any necessary guest agents or drivers.

Virtual Machines Components

1. Operating System

- A Virtual Machine is an Operating System or application environment that runs on software that replicates special hardware.
- The end-user experience when utilizing a virtual machine is identical to that of special hardware.



Virtual Machines Components

2. VMWare Tools

- VMware Tools is a set of services and modules that enable several features in VMware products for better management of guests operating systems and seamless user interactions with them.

VMware Tools has the ability to:

- Pass messages from the host operating system to the guest operating system.
- Customize guest operating systems as a part of the vCenter Server and other VMware products.
- Run scripts that help automate guest operating system operations.
- The scripts run when the power state of the virtual machine changes.
- Synchronize the time in the guest operating system with the time on the host operating system.

Virtual Machines Components

General

Guest OS: Other 2.6.x Linux (64-bit)
VM Version: vmx-10
CPU: 1 vCPU
Memory: 1024 MB
Memory Overhead: 26,04 MB

VMware Tools: ? Running (3rd-party/Independent)
IP Addresses: [View all](#)

DNS Name: Debian

State: Powered On
Host:

Active Tasks:

vSphere HA Protection: ? N/A

Player ▾ || ▾ □ □

- File ▸
- Power ▸
- Removable Devices ▸
- Send Ctrl+Alt+Del
- Manage ▸
 - Install VMware Tools...
 - Message Log
 - Virtual Machine Settings... Ctrl+D
- Full Screen Ctrl+Alt+Enter
- Unity
- Help ▸
- Exit

Hardware Options

Device	Summary
Memory	2 GB
Processors	2
Hard Disk (NVMe)	60 GB
CD/DVD 2 (SATA)	Using file D:\W2.UTIL\W3.Vm...
CD/DVD (SATA)	Auto detect
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

Virtual Machines Components

3. Virtual Resources and Hardware

Processors

- A processor is a device that processes programmed instructions. The more processors that user assigns to a logical partition, the greater the number of concurrent operations that the logical partition can run at any given time.

Memory

- Processors use memory to temporarily hold information. Memory requirements for logical partitions depend on the logical partition configuration, I/O resources assigned, and applications used.
- Terminal and console options for logical partitions

I/O devices

- I/O devices allow user managed system to gather, store, and transmit data. I/O devices are found in the server unit itself and in expansion units that are attached to the server.
- I/O devices can be embedded into the unit, or they can be installed into physical slots.

Virtual Machine – Components

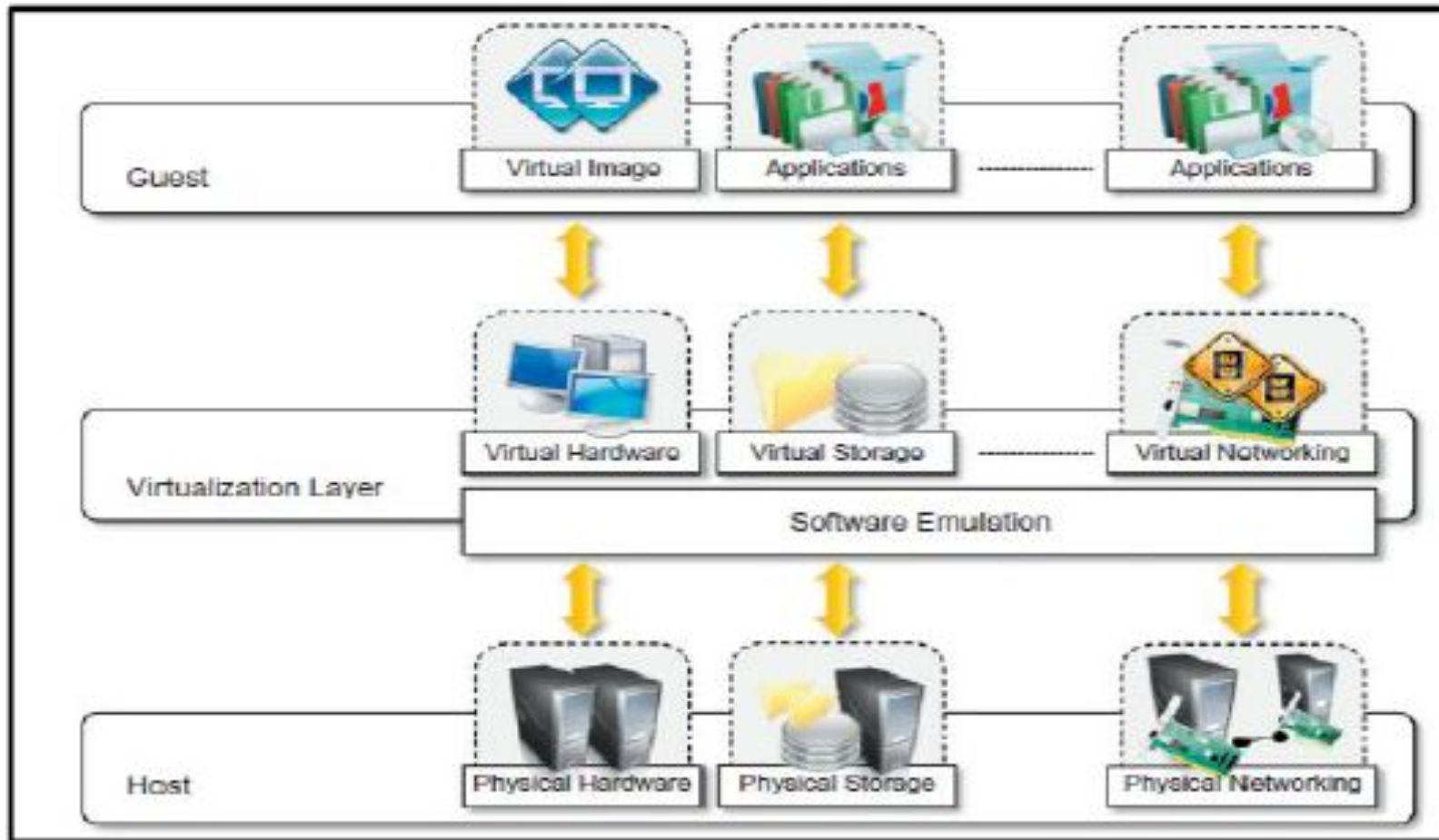
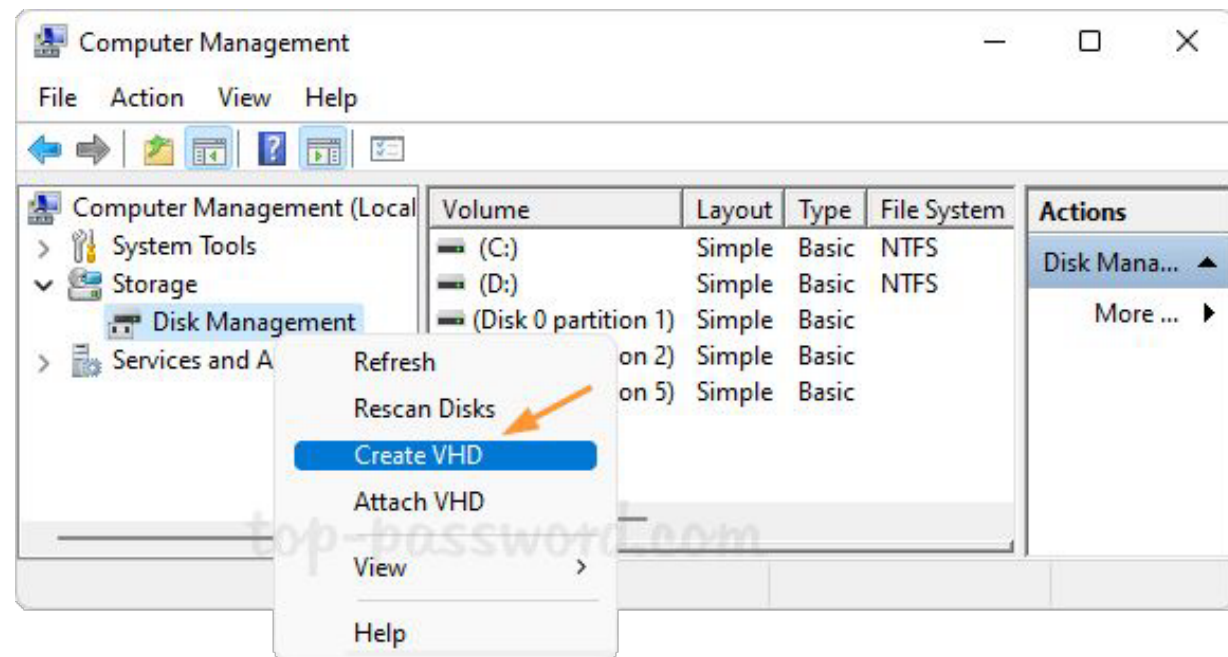


Figure Virtualized Environment

Virtual Hard Disk (VHD)

- A virtual hard disk (VHD) is a file format that acts like a physical hard drive and is often used with virtual machines (VMs).
- VHDs can contain disk partitions, file systems, files, and folders, and can be used for storage, system or data drives, or archival purposes.
- They can also be used with physical machines



Virtual Hard Disk (VHD) – Example

- Some of the common virtual hard disk examples shown
 - Virtual Disk Image (VDI), used by Oracle VirtualBox
 - Virtual Machine Disk (VMDK), used by VMware
 - Virtual Hard Disk (VHD), used by Microsoft and Citrix
 - Virtual Hard Disk (VHDX), used by Microsoft



citrix™



Virtual Hard Disk (VHD) Uses in Virtualization

Operate on various host environments

- VHDs allow VMs to operate on different host environments, which can facilitate easy storage management.

Support dynamic expansion and contraction

- VHDs support the dynamic expansion and contraction of disk storage.

Optimize hardware resources

- VHDs can optimize the use of hardware resources like CPU, memory, and storage.

Virtual Hard Disk (VHD) Uses in Virtualization

Enhance system security and scalability

- VHDs can enhance system security and scalability, which can help reduce operational costs and improve business continuity.

Reset the environment

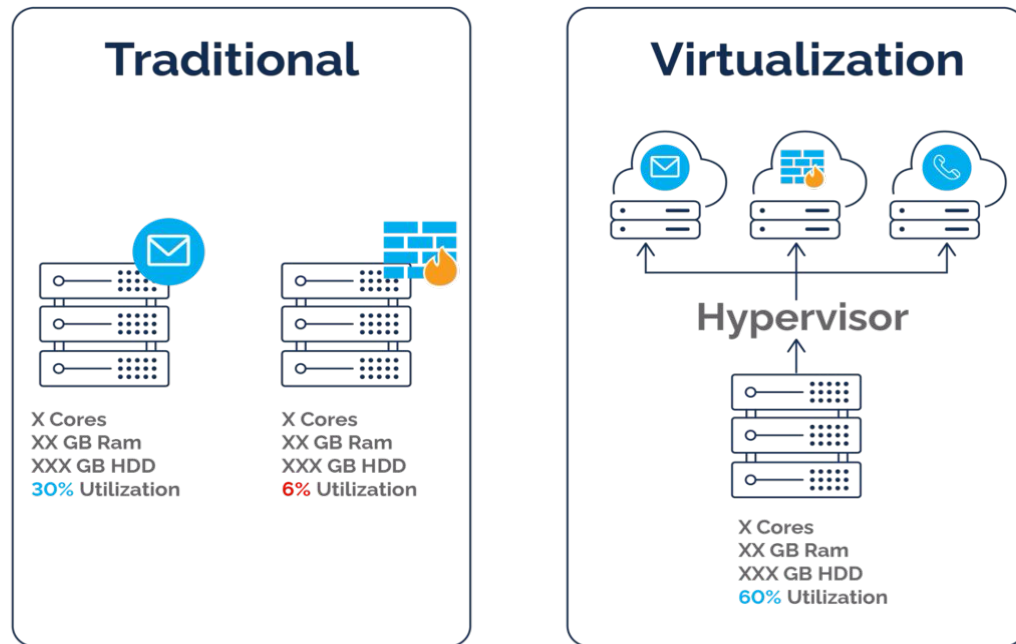
- VHDs can be ideal for resetting the environment to an initial state before malformed data is sent.

Hide large quantities of data

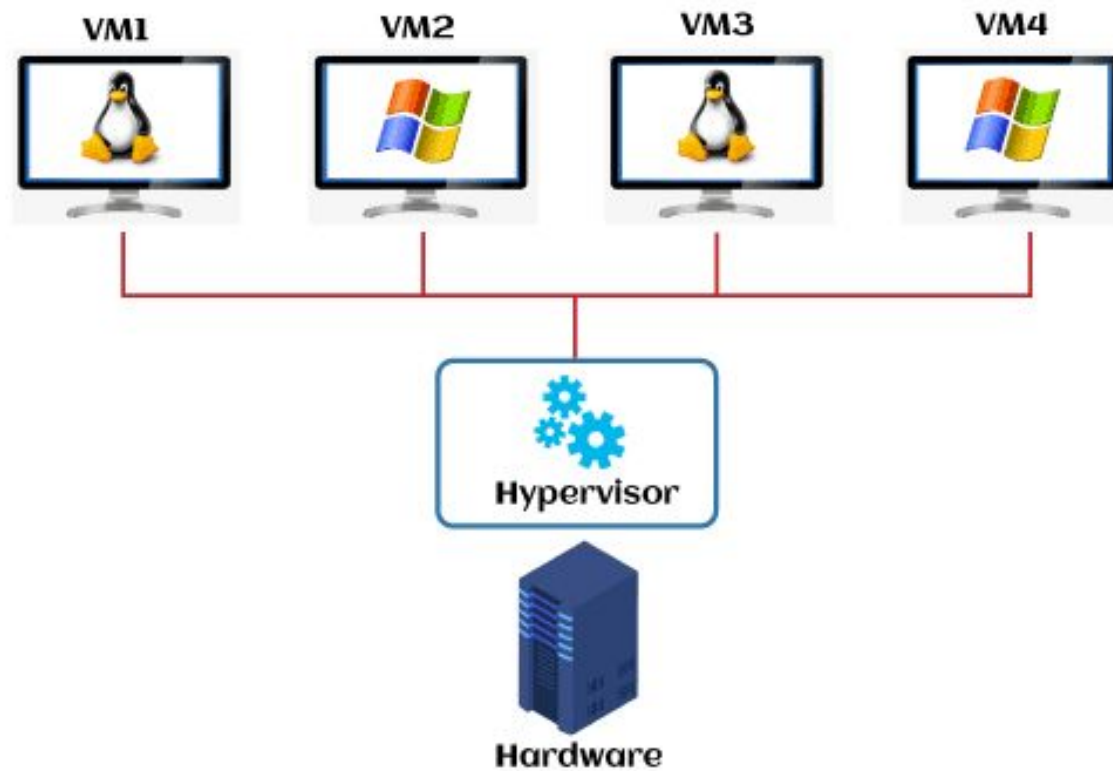
- VHDs can be used to hide large quantities of data because VMs dynamically change in size while in use, which can circumvent traditional data integrity programs.

Hypervisor

- A hypervisor is a software layer that allows user to run multiple virtual machines (VMs) on a single physical machine.
- It's also known as a virtual machine monitor (VMM).
- Hypervisors isolate the hypervisor's operating system and resources from the VMs, and then allocate physical computing resources like CPU and memory to each VM as needed



Hypervisor

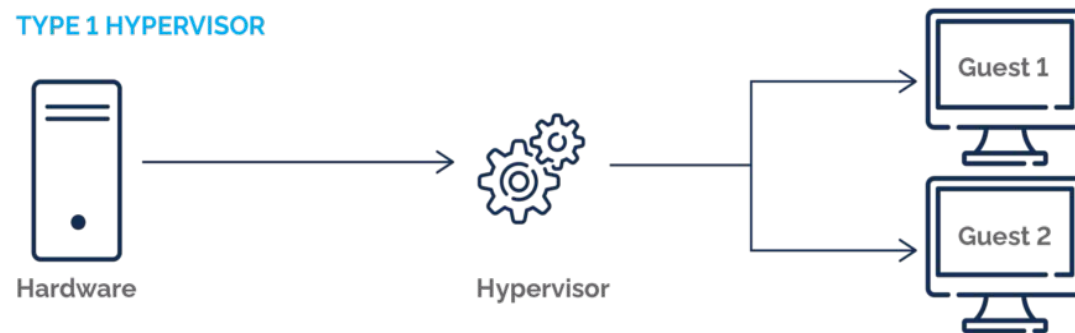


Types of Hypervisor

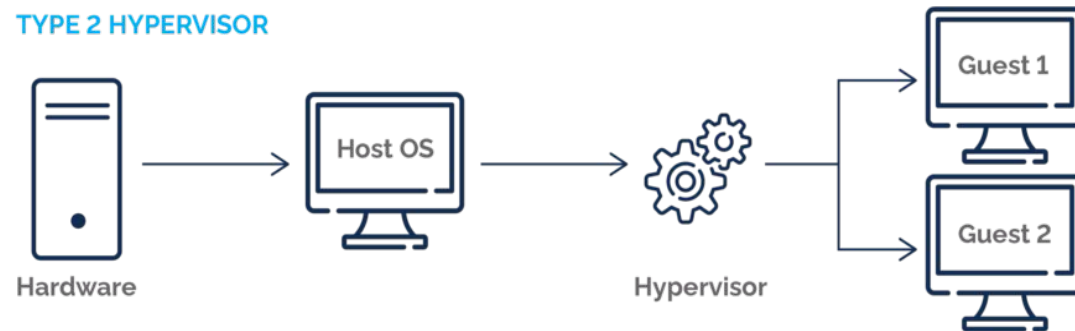
- The following types it has
- Type 1
- Type 2

HYPERVISOR TYPES

TYPE 1 HYPERVISOR



TYPE 2 HYPERVISOR



Types of Hypervisor

Type 1 Hypervisor

- Most common in enterprise data centers, a type 1 hypervisor replaces the host's operating system and lies right on top of the hardware. For this reason, type 1 hypervisors are also called bare metal hypervisors or embedded hypervisors.

Type 1 Hypervisor Examples

Here's a list of hypervisors of the type 1 variety:

- VMware hypervisors like vSphere, ESXi and ESX
- Microsoft Hyper-V
- Oracle VM Server
- Citrix Hypervisor

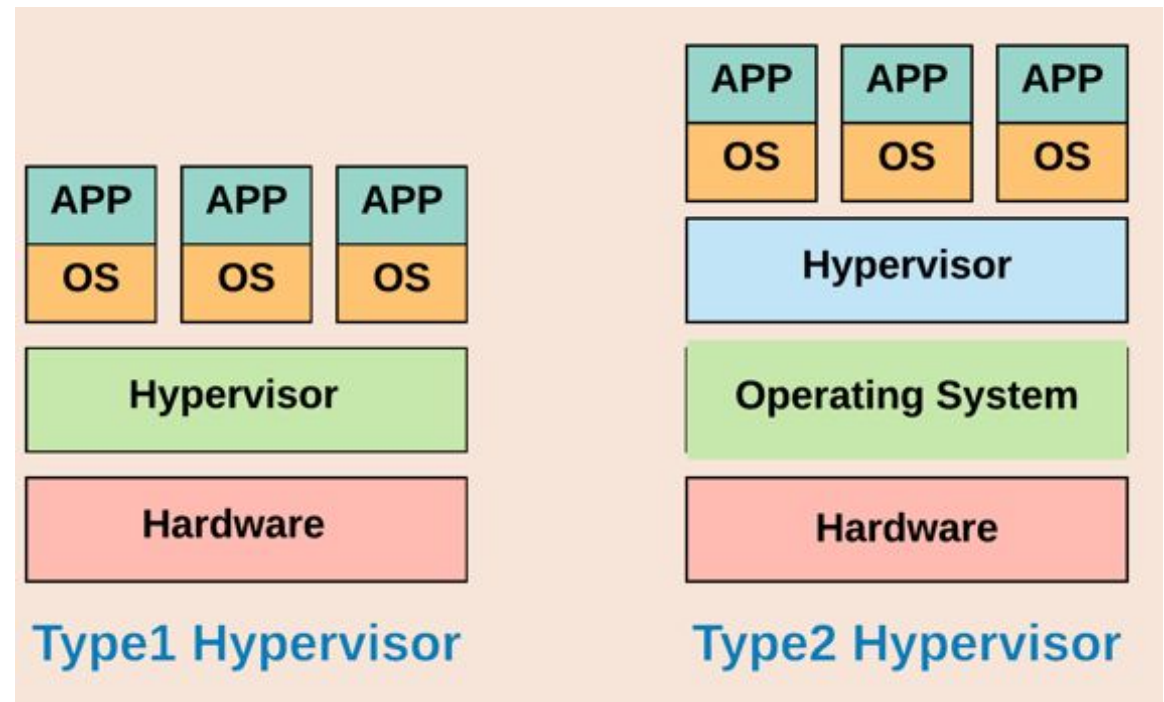
Types of Hypervisor

Type 2 Hypervisor

- A type 2 hypervisor is hosted, running as software on the O/S, which in turn runs on the physical hardware. This form of hypervisor is typically used to run multiple operating systems on one personal computer, such as to enable the user to boot into either Windows or Linux.

Type 2 Hypervisor Examples

- Type 2 hypervisor examples include:
 - VMware Workstation
 - VMware Fusion
 - Oracle VirtualBox
 - Oracle Solaris Zones
 - Oracle VM Server for x86



Hypervisor Management Applications

Hypervisor Management Applications

- VMware ESXi (an enterprise-environment, type-1 hypervisor developed by VMware, a subsidiary of Broadcom, for deploying and serving virtual computers)
- VirtualBox (Free and type 1 hypervisor developed by Oracle)
- Hyper-V (It is a robust Type 1 hypervisor that integrates seamlessly with Windows environments.)
- VMware Fusion (VMware Fusion is a hypervisor solution similar to Workstation Pro but created for macOS users.)
- Xen (It is an open-source type 1 hypervisor developed by the Xen Project)
- Virtual PC (PowerPC Mac hosts and a hypervisor for Microsoft Windows hosts)
- VMware Workstation (x64 versions of Windows and Linux)

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THANK YOU